

Skill-14

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himasree@LAPTOP-41COT5AA:~/OSLab$ nano append_stderr.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc append_stderr.c -o append_stderr
himasree@LAPTOP-41COT5AA:~/OSLab$ echo "Original line" > append.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ cat append.txt
Original line
himasree@LAPTOP-41COT5AA:~/OSLab$ ./append_stderr
=====
APPEND & STDERR REDIRECTION
=====

Enter append output file: append.txt
Enter stderr output file: errors.txt

=====
Redirection tests completed successfully.
=====
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ cat errors.txt
Error: File could not be opened.
Error: Demonstrating stderr redirection.
Error: This output is captured in the error file.
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ cat append.txt
Original line
New line appended to the file.
Existing data was preserved.
New line appended to the file.
Existing data was preserved.
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```

```

himasree@LAPTOP-41COT5AA:~/OSLab$ cat append_stderr.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/wait.h>

#define BUFFER_SIZE 1024

/* -----
APPEND REDIRECTION: >>
----- */
void append_redirection(const char *filename)
{
    int fd = open(filename, O_WRONLY | O_CREAT | O_APPEND, 0644);

    if (fd < 0)
    {
        perror("Error opening append file");
        return;
    }

    if (dup2(fd, STDOUT_FILENO) < 0)
    {
        perror("dup2 failed");
        close(fd);
        return;
    }
}

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        close(fd);

        printf("New line appended to the file.\n");
        printf("Existing data was preserved.\n");
    }

/* -----
STDERR REDIRECTION: 2>
----- */
void stderr_redirection(const char *filename)
{
    int fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0)
    {
        perror("Error creating stderr file");
        return;
    }

    if (dup2(fd, STDERR_FILENO) < 0)
    {
        perror("dup2 failed");
        close(fd);
        return;
    }

    close(fd);

    /* These messages go into the stderr file */
    fprintf(stderr, "Error: File could not be opened.\n");
}

```

```

int main()
{
    char append_file[100];
    char error_file[100];

    printf("=====\n");
    printf("      APPEND & STDERR REDIRECTION\n");
    printf("=====\n");

    printf("\nEnter append output file: ");
    scanf("%99s", append_file);

    printf("Enter stderr output file: ");
    scanf("%99s", error_file);

    /* -----
       APPEND REDIRECTION
       ----- */

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {

```

```

        wait(NULL);

        /* -----
           STDERR REDIRECTION
           ----- */

        pid = fork();

        if (pid < 0)
        {
            perror("fork failed");
            return 1;
        }

        if (pid == 0)
        {
            stderr_redirection(error_file);
            exit(0);
        }

        wait(NULL);

        printf("\n=====\n");
        printf("Redirection tests completed successfully.\n");
        printf("=====\n");

        return 0;
    }
}
himasree@LAPTOP-41COT5AA:~/OSLab$ |

```